



U.S CHRONIC DISEASE INDICATORS ANALYSIS

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INTRODUCTION

Chronic diseases such as **heart disease, cancer, diabetes, asthma, and obesity** are among the leading causes of illness and mortality in the United States. Monitoring these diseases is essential for understanding public health trends and supporting evidence-based healthcare decisions.

This project analyses the **U.S. Chronic Disease Indicators (CDI)** dataset using **Python for Data Analysis**. The dataset contains **283,950 records** collected from multiple U.S. states between **2020 and 2023**, covering various chronic disease indicators, demographic groups, and health measures.

Using **Pandas, NumPy, Matplotlib, and Seaborn**, the project performs data preprocessing, exploratory data analysis (EDA), statistical analysis, feature engineering, and data visualization to identify meaningful patterns, trends, and insights that can support public health planning and decision-making





PROBLEM STATEMENT

- CHRONIC DISEASES ARE ONE OF THE LEADING CAUSES OF ILLNESS AND MORTALITY IN THE UNITED STATES. THE **U.S. CHRONIC DISEASE INDICATORS (CDI)** DATASET CONTAINS A LARGE VOLUME OF HEALTH DATA COLLECTED ACROSS MULTIPLE STATES, YEARS, AND DEMOGRAPHIC GROUPS. ANALYSING THIS COMPLEX DATASET MANUALLY MAKES IT DIFFICULT TO IDENTIFY DISEASE TRENDS, COMPARE HEALTH INDICATORS, AND SUPPORT EFFECTIVE PUBLIC HEALTH DECISION-MAKING.
- THIS PROJECT APPLIES **PYTHON-BASED DATA ANALYSIS TECHNIQUES** TO TRANSFORM RAW DATA INTO MEANINGFUL INSIGHTS THROUGH DATA PREPROCESSING, STATISTICAL ANALYSIS, AND VISUALIZATION.
- **KEY CHALLENGES**
 - LARGE DATASET WITH **283,950 RECORDS**
 - MULTIPLE HEALTH TOPICS AND DISEASE INDICATORS
 - MISSING VALUES AND INCONSISTENT RECORDS
 - DIFFICULT TO IDENTIFY TRENDS THROUGH MANUAL ANALYSIS
 - NEED FOR DATA-DRIVEN INSIGHTS TO SUPPORT PUBLIC HEALTH DECISIONS

PROJECT OBJECTIVES

THE PRIMARY OBJECTIVE OF THIS PROJECT IS TO ANALYSE THE **U.S. CHRONIC DISEASE INDICATORS (CDI)** DATASET USING PYTHON AND GENERATE MEANINGFUL INSIGHTS THROUGH DATA ANALYSIS AND VISUALIZATION.



OBJECTIVES

- ANALYSE CHRONIC DISEASE INDICATORS ACROSS DIFFERENT **U.S. STATES** AND **TIME PERIODS (2020–2023)**.
- PERFORM **DATA PREPROCESSING** BY HANDLING MISSING VALUES, DUPLICATES, AND DATA TYPE INCONSISTENCIES.
- CONDUCT **EXPLORATORY DATA ANALYSIS (EDA)** TO UNDERSTAND THE DISTRIBUTION AND CHARACTERISTICS OF THE DATASET.
- APPLY **STATISTICAL ANALYSIS** TO MEASURE TRENDS, VARIABILITY, AND RELATIONSHIPS AMONG HEALTH INDICATORS.
- CREATE **INTERACTIVE AND INFORMATIVE VISUALIZATIONS** TO IDENTIFY PATTERNS AND COMPARE DISEASE INDICATORS.
- GENERATE **DATA-DRIVEN INSIGHTS AND RECOMMENDATIONS** TO SUPPORT PUBLIC HEALTH DECISION-MAKING.

DATASET OVERVIEW

Attribute	Details
Dataset Name	U.S. Chronic Disease Indicators (CDI)
Source	Centers for Disease Control and Prevention (CDC)
Total Records	283,950
Total Columns	24
Time Period	2020 – 2023
Data Type	Numerical & Categorical

- **Key Attributes**
- **Year Start & Yearend** – Reporting period
- **Locations** – U.S. State or Location
- **Topic** – Disease category
- **Question** – Health indicator
- **Data Value** – Measured value
- **DataValueType** – Type of measurement
- **LowConfidenceLimit & HighConfidenceLimit** – Confidence interval
- **StratificationCategory1** – Demographic category

PROJECT METHODOLOGY

Dataset Collection

Data Preprocessing

Data Transformation

Feature Engineering

Exploratory Data Analysis (EDA)

Statistical Analysis

Data Visualization

Key Insights

Business Recommendations

conclusion





DATA PREPROCESSING

- **Data Cleaning**

- Removed duplicate records.
- Handled missing values.
- Converted data types.
- Renamed columns for improved readability.

- **Data Transformation**

- Standardized categorical variables.
- Extracted Latitude and Longitude from geolocation data.
- Optimized data types for memory efficiency.
- Enhanced data consistency for reliable analysis.



EXPLORATORY DATA ANALYSIS (EDA)

- Dataset Structure Analysis
- Missing Value and Duplicate Analysis
- Descriptive Statistical Analysis
- Disease, State, and Data Source Analysis
- Year-wise and Disease-wise Trend Analysis
- Grouped Analysis for Pattern Identification



FEATURE ENGINEERING

- Created **Region Type** for geographical classification.
- Developed **Survey Type** based on data sources.
- Calculated **Confidence Width** to measure estimate precision.
- Created **Demographic Group** for simplified demographic analysis.



STATISTICAL ANALYSIS

- **Measures of Dispersion**
 - Mean
 - Variance
 - Standard Deviation
 - Coefficient of Variation
- **Distribution Analysis**
 - Skewness
 - Kurtosis
- **Correlation Analysis**
 - Correlation Matrix
 - Heatmap



DATA VISUALIZATION

- Histogram – Disease Value Distribution
- Box Plot – Outlier Detection
- Bar Chart – State-wise Record Distribution
- Pie Chart – Disease Category Distribution
- Bar Plot – Year-wise Disease Comparison
- Correlation Heatmap – Numerical Variable Correlation
- Grouped Bar Chart – Year × Disease Comparison



KEY INSIGHTS

- Chronic disease records vary across U.S. states and territories.
- Disease prevalence differs across health topics and demographic groups.
- Year-wise analysis reveals changing disease indicator trends.
- Outlier detection identified extreme disease indicator values.
- Correlation analysis showed strong relationships between disease values and confidence limits.



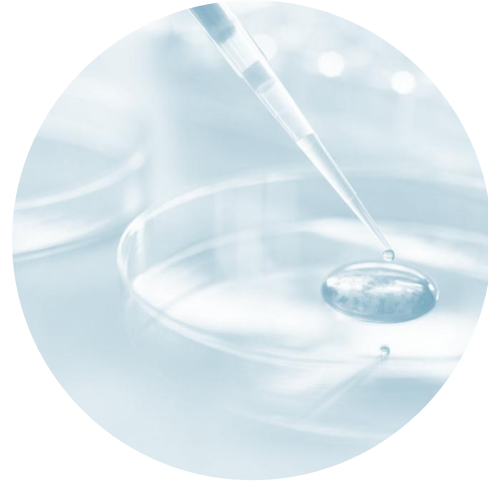
BUSINESS RECOMMENDATIONS

- **Prioritize high-risk regions** by allocating healthcare resources to states with higher disease indicator values.
- **Strengthen preventive healthcare programs** through regular screening, early diagnosis, and awareness campaigns.
- **Develop targeted interventions** for vulnerable demographic groups based on age, sex, and race/ethnicity.
- **Improve data quality and reporting** by ensuring consistent and complete public health data collection across all regions.
- **Use data-driven decision-making** to monitor disease trends and support effective public health policy and resource planning.

CONCLUSION

- SUCCESSFULLY ANALYSED THE U.S. CHRONIC DISEASE INDICATORS DATASET USING PYTHON.
- PERFORMED DATA PREPROCESSING, FEATURE ENGINEERING, EXPLORATORY DATA ANALYSIS, STATISTICAL ANALYSIS, AND VISUALIZATION TO IDENTIFY MEANINGFUL PATTERNS.
- THE ANALYSIS REVEALED VARIATIONS IN DISEASE INDICATORS ACROSS YEARS, STATES, AND DEMOGRAPHIC GROUPS.
- THE INSIGHTS OBTAINED CAN SUPPORT EVIDENCE-BASED PUBLIC HEALTH PLANNING AND DATA-DRIVEN DECISION-MAKING.





THANK YOU

